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# Data Analysis Essentials

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# Lab Practice

Practice:

Week 3: Expanding and Squeezing a NumPy Array

a) Expanding a NumPy array

b) Squeezing a NumPy array

c) Sorting in NumPy Arrays

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# ndarray

## a) Expanding a NumPy Array

Expanding a NumPy array means **adding a new axis (dimension)** to an existing array. This is useful for machine learning, image processing

np.expand_dims(array, axis)

- **array** → Input array
- **axis** → Position where new dimension is inserted

# Expand 1D Array into Row Vector

**a) Expanding a NumPy Array**

```
import numpy as np
arr = np.array([10,20,30,40])
new_arr = np.expand_dims(arr, axis=0)
print(new_arr)
print("Shape:", new_arr.shape)
```

**Output**

```
[[10 20 30 40]]
Shape: (1, 4)
```

# Expand into Column Vector

**Program**

```
import numpy as np
arr = np.array([10,20,30,40])
new_arr = np.expand_dims(arr, axis=1)
print(new_arr)
print("Shape:", new_arr.shape)
```

**Output**

```
[[10]
 [20]
 [30]
 [40]]

Shape: (4,1)
```

# Expand a 2D Array

**Program**

```
import numpy as np
arr = np.array([[1,2],
                [3,4]])
new_arr = np.expand_dims(arr, axis=0)
print(new_arr)
print("Shape:", new_arr.shape)
```

**Output**

```
(2, 2)
[[[1 2]
[3 4]]]
Shape: (1, 2, 2)
```

# Expand a 2×3 matrix into a 3D array

**Program** It adds a new dimension at the end of the array.

```
import numpy as np
matrix = np.array([[1,2,3], [4,5,6]])
result = np.expand_dims(matrix, axis=2)
print(result)
print(result.shape)
```

**Output**

```
[[[1]
[2]
[3]]
[[4]
[5]
[6]]]
(2, 3, 1)
```

# Squeezing a NumPy Array

Squeezing removes dimensions of size 1 from an array.

**Program**

```
import numpy as np
arr = np.array([[[10,20,30]]])
print("Original Shape:", arr.shape)
new_arr = np.squeeze(arr)
print(new_arr)
print("New Shape:", new_arr.shape)
```

**Output**

```
Original Shape: (1, 1, 3)
[10 20 30]
New Shape: (3,)
```

# Squeezing a NumPy Array

**Program**

```
import numpy as np
arr = np.array([[10],
                [20],
                [30]])

print(arr.shape)
result = np.squeeze(arr)
print(result)
print(result.shape)
```

**Output**

```
(3, 1)
[10 20 30]
(3,)
```

# Squeezing a NumPy Array

Squeeze only the second dimension.

**Program**

```
import numpy as np
arr = np.array([[[10,20,30]]])
print(arr.shape)
result = np.squeeze(arr, axis=1)
print(result)
print(result.shape)
```

**Output**

```
(1, 1, 3)
[[10 20 30]]
(1, 3)
```

# Sorting in NumPy Arrays

Sorting arranges elements in ascending order by default.

**Syntax**

np.sort(array)

**Program**

```
import numpy as np
arr = np.array([25,12,45,10,8])
sorted_arr = np.sort(arr)
print(sorted_arr)
```

**Output**

[ 8 10 12 25 45]

# Sort each row

Rows are sorted independently

**Program**

```
import numpy as np
arr = np.array([[9,4,7],
                [6,2,8]])
print(np.sort(arr))
```

**Output**

```
[[4 7 9]
[2 6 8]]
```

# Sort column wise

Each column is sorted separately.

**Program**

```
import numpy as np
arr = np.array([[9,4,7],
                [6,2,8]])
result = np.sort(arr, axis=0)
print(result)
```

**Output**

```
[[6 2 7]
[9 4 8]]
```

# Descending Order

**Program**

```
import numpy as np
arr = np.array([45,20,70,10])
descending = np.sort(arr)[::-1]
print(descending)
```

**Output**

```
[70 45 20 10]
```

# Sort Strings

**Program**

```
import numpy as np
names = np.array(["Ravi","Anil","Sita","Kiran"])
print(np.sort(names))
```

**Output**

```
['Anil' 'Kiran' 'Ravi' 'Sita']
```

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# Find the largest and smallest values after sorting

**Program**

```
import numpy as np
arr = np.array([45,12,78,33,90])
sorted_arr = np.sort(arr)
print("Smallest:", sorted_arr[0])
print("Largest :", sorted_arr[-1])
```

**Output**

```
Smallest: 12
Largest : 90
```

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# Thank You